

● EMOUVA

How Emouva works.

A technical overview of the AI agent that researches, forms falsifiable theses, and invests like a disciplined value investor — with the human holding the mandate.

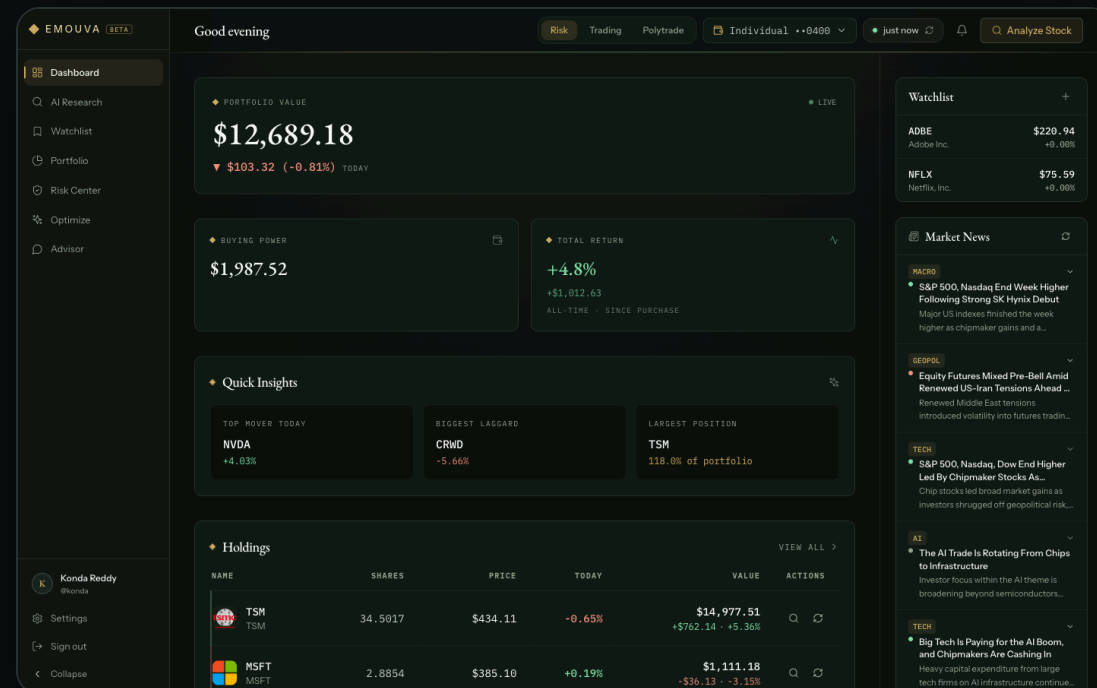
TECHNICAL OVERVIEW · v1 · emouva.com

[Trading agent](#) · [Central intelligence](#) · [Stock research](#) · [Themes](#) · [Community](#)

◆ WHAT EMOUVA IS

An AI **partner**, not another trading app.

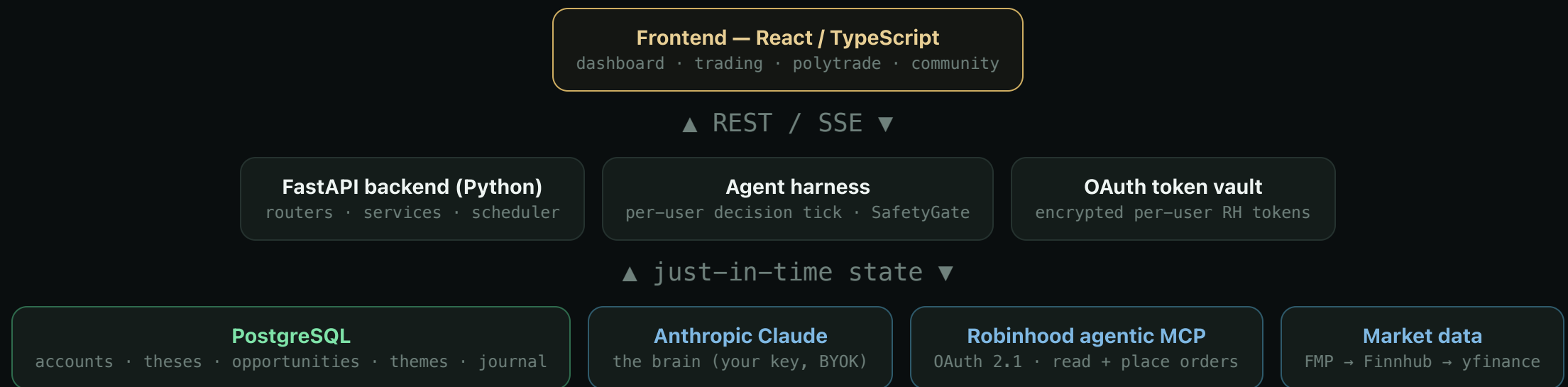
- **You set the mandate** — goal, risk limits, and the dollar threshold above which it must ask you first.
- **It runs a disciplined loop** — screens the market on a cadence, buys rarely, and explains every move (and every pass).
- **Value-investing brain** — fair value, margin of safety, a written thesis with machine-checkable exit rules.
- **Local-first & open source** — you run it on your machine with your own keys; emouva.com is the community.



The dashboard – portfolio, holdings, insights & market news

◆ SYSTEM ARCHITECTURE

One backend, many isolated agents.



The **context window is scratch space, not storage**. Every decision tick cold-starts a small, cache-friendly context; durable state lives in Postgres and is pulled in just-in-time — so cost per tick stays bounded no matter how long an agent has run.

◆ THE TRADING AGENT · "THE PARTNER"

Every tick is one cheap decision.

01 · SYNC

Reconcile

Pull the live account from Robinhood; idle-skip if nothing to do.

02 · WATCH

Check theses

Evaluate each holding's machine-checkable falsifiers on fresh data.

03 · GUARD

Drawdown backstop

Any holding down past the limit forces a thesis re-review.

04 · SHORTLIST

Candidate menu

Only names clearing margin-of-safety; falling knives excluded.

05 · DECIDE

One brain call

Buy / sell / **hold** with rationale. HOLD is the default.

06 · VERIFY

Double-check

A web-search sanity pass before any auto-execute buy.

07 · GATE

Execute

Through the SafetyGate → Robinhood, or queue for approval.

"Most days it does nothing — that is the point." The agent hunts for reasons **not** to act; clearing the hurdle makes a name eligible, not a buy. It only acts on a clearly compelling, high-conviction standout that beats holding cash — or its weakest current holding.

◆ SAFETY — HARD CODE, NOT A PROMPT

The agent proposes. **You approve.**

Kill switch

One flag halts every tick and blocks all placement, platform-wide.

Allowlist

Orders are bound to a vetted symbol set — nothing off-list is placed.

Spend caps

Per-trade, daily and total dollar ceilings enforced before any order.

Concentration limit

Hard max position weight & cash floor — cannot be crossed, even with your approval.

Approval threshold

Any order over your dollar limit queues for a human yes/no; below it, auto-executes.

Fenced funds

Live trades run only inside Robinhood's separately-funded agentic account.

Fail-closed: if any gate can't be verified, nothing is sent. Robinhood's own approval is soft/prompt-driven — so the gate is a hard code layer (backend/app/harness/safety.py), never a model instruction. Every proposal, decision and fill is written to an append-only journal.

◆ CENTRAL INTELLIGENCE · THE OPPORTUNITY POOL

Research once. Reuse everywhere.

~250

SEEDED UNIVERSE

Mega-caps, S&P, top US-traded ADRs,
plus daily gainers/losers scans.

Stats gate

CHEAP FILTER · NO LLM

Reject revenue-downtrend / unprofitable
value-traps before spending a token.

Analysis

LLM HARNESS

Thesis, growth, falsifiers & a 4-lens red-
team for every survivor.

1 · 2 · 0

CATEGORY

1 tradeable & understood · 2 looks good,
not understood · 0 rejected.

A **shared, pre-reasoned directory** of stocks — computed once by the harness and reused by every user's agent, the morning screen, and Themes (no per-account LLM re-analysis). Refresh is split by what actually changes: fundamentals are quarterly, prices daily, the thesis stays stable. Admins curate it from a private console.

Value first — price second.

- **Fair-value triangulation** — a range from DCF + analyst targets + a conservative earnings multiple; flagged *not confident* when methods disagree.
- **Margin of safety** — a name is only eligible when price sits a user-set % below conservative fair value (default 30%).
- **Circle of competence** — sector include/exclude; "if it's outside the circle, the price doesn't matter."
- **Quality bias** — gross profitability, low leverage, stable margins; junk rallies are rented, not owned.

DCF discounted cash flow

FCF-based, currency-correct (FX-converted), deliberately conservative.

+ analyst consensus

Forward price-target consensus where the data provider serves it.

+ earnings multiple

A conservative P/E-anchored cross-check on the range.

= a fair-value range

Conservative = mean $\times$ 0.9; feeds the margin gate, the brain, and the screen.

Every position is a **falsifiable** bet.

Arm on entry

On any buy, the harness writes the thesis and arms machine-checkable falsifiers — {metric, comparator, threshold}.

Red-team

A 4-lens attack — accounting, moat, leverage, valuation. A buy auto-proceeds only if it survives ≥ 3 .

Daily sweep

Falsifiers are re-evaluated on fresh data. A trip triggers a harness review — reaffirm or propose an exit.

Never auto-sells

Exits route through the same gate. A broken thesis is distilled into a lesson → a new principle.

🔴 Falling-knife guard

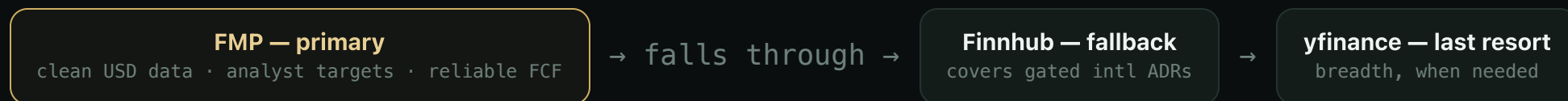
Pure-technical trend scoring (SMA/returns/RSI) **hard-excludes** names in steep decline from the buy menu — however large the margin. "Don't catch a falling knife."

⚠️ Catastrophic review

A holding down past a wide threshold from cost forces a fresh-data thesis re-review — *not* a stop-loss. A big paper loss is never itself a sell reason; a broken thesis is.

◆ MARKET-DATA LAYER

Resilient by design.

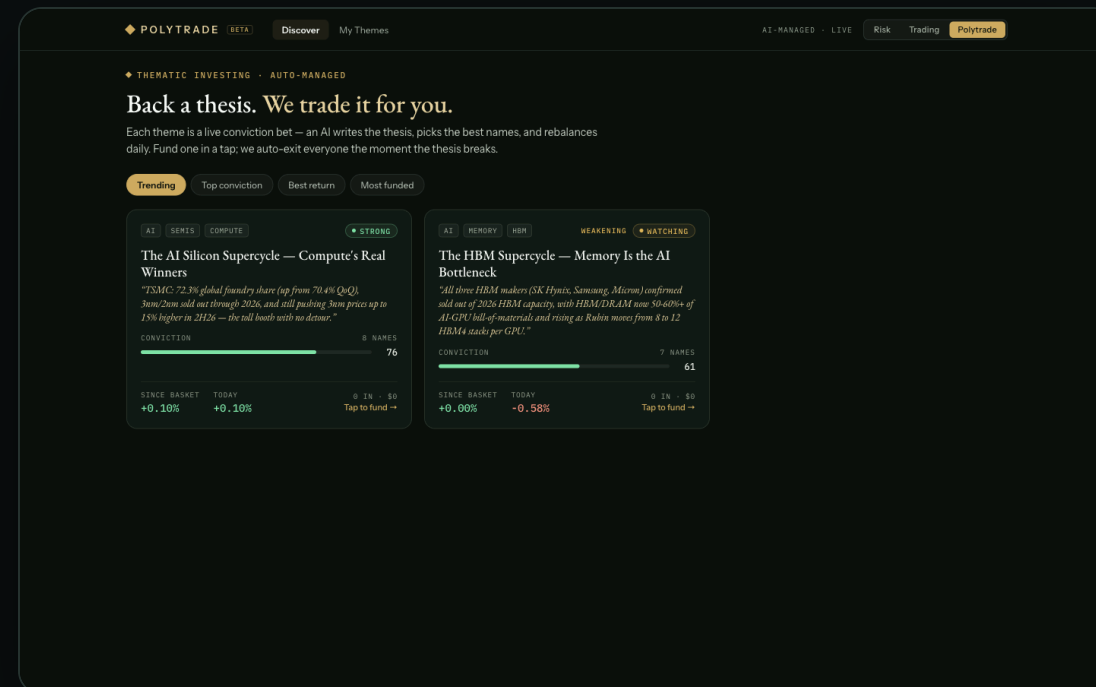


- **One normalized shape** — every provider maps to identical quote / fundamentals / history so nothing downstream changes when a vendor is reordered or dropped.
- **A single env knob** — MARKET_PROVIDER_ORDER reorders the chain; per-provider rate-limiting + caching keep it fast and within plan caps.
- **Correctness fixes baked in** — FX-converted free cash flow for foreign filers, so DCFs aren't inflated by the exchange rate.

Back a thesis. We trade the basket.

AI-curated thematic baskets ("HBM supercycle", "AI silicon") you fund in one tap — auto-managed to target weights, monitored daily, and auto-exited the moment the thesis breaks.

- **3-stage constituent pipeline** — AI proposes a universe (web search) → we enrich each with our own metrics → AI final-selects 4–8 with capped weights.
- **Theme-level Living Thesis** — same falsifiers + red-team, applied to the whole basket.
- **Version-driven reconciler** — crash-safe, idempotent propagation of target changes across every holder.
- **Managed-downside metric** — estimates the loss if the thesis weakens and we exit early vs. left alone.



Polytrade — conviction-weighted, auto-managed baskets

◆ LOCAL-FIRST · BYOK · COMMUNITY

Your machine, your keys, one community.

Run it locally

Brokerages don't grant agent access to arbitrary sites — so you run the app yourself. Robinhood OAuth redirects to your own `localhost:8001`; tokens never leave your box.

Bring your own key

The AI runs on **your** Anthropic key, entered in Settings and sent per-request — never stored on a server. No shared bill, no lock-in.

Share on emouva.com

One public channel: post a one-tap "Wrapped"-style P&L card, swap tricks, and learn how others run their agents.

Paper first, always: the same brain runs on simulated money before a single real order. The result is a community of AI traders comparing disciplined, thesis-driven results — not a casino.

◆ THE STACK · AT A GLANCE

Boring where it counts.

Backend

Python · FastAPI · SQLAlchemy (async) · PostgreSQL · APScheduler. The agent harness is a decision-tick brain over the Anthropic Messages API with a forced decision tool.

Frontend

React · TypeScript · Vite. Four surfaces — Risk (analytics), Trading (the Partner), Polytrade (themes), Community.

Integrations

Robinhood agentic MCP over OAuth 2.1 (DCR + PKCE, encrypted token vault) · Anthropic Claude · FMP / Finnhub / yfinance.

Core tables

accounts · account_positions · opportunities · theses · themes · theme_allocations · community_posts

Value investing

Falsifiable theses

Fail-closed safety

Compute-once research

Local-first · BYOK

Open source (MIT)

Emouva — screen everything, buy rarely, explain every move · emouva.com · github.com/kondareddy-th/emouva